

# Bhoomika Goel

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## SUMMARY

AI/ML Engineer with hands-on experience building production-style LLM systems, federated learning frameworks, and deep learning pipelines. Skilled in LangChain, PyTorch, FastAPI, and SHAP-based explainability. Currently building an agentic research system with multi-agent coordination using LangGraph.

## PROJECTS

### ResearchMind AI :Autonomous Multi-Agent Research Intelligence System

Python, Gemini API, Groq, ChromaDB, Sentence Transformers, Agentic AI | 2026 | GitHub

- Architected an autonomous multi-agent AI system comprising Planner, Research, Summarizer, Synthesis, Critic, and Formatter agents, enabling task decomposition, retrieval, cross-document reasoning, self-evaluation, and research report generation.
- Designed a stateful semantic memory architecture using vector embeddings and ChromaDB, supporting similarity-based retrieval, adaptive knowledge reuse, planner consistency, and reduced redundant LLM inference across sessions.
- Implemented advanced agentic AI and systems engineering patterns including multi-agent orchestration, critique-driven refinement, multi-source retrieval, semantic ranking, caching, retry/fallback routing, and observable stateful workflows for reliable end-to-end research automation.

### TriMind AI: Advanced & Scalable RAG Pipeline with Intelligent Retrieval, Query Optimization, Memory & Evaluation

Python, LangChain, Pinecone, Gemini API, FastAPI | 2026 | GitHub

- Built a modular Retrieval-Augmented Generation (RAG) pipeline with multi-document retrieval, query optimization, and conversational memory, enabling context-aware, source-grounded responses across large document sets.
- Designed an evaluation-driven system with retrieval logging, grounding and confidence scoring, and hallucination detection to improve response reliability and transparency.
- Developed a FastAPI-based backend with caching and structured logging, integrating LangChain, Pinecone, and Gemini API for modular, production-style AI querying.

### Federated Learning-Based Parkinson Disease Prediction System

Python, Scikit-learn, SHAP, Pandas, NumPy | 2026 | GitHub

- Built a privacy-preserving multi-round federated learning framework across 5 heterogeneous clients for Parkinson's disease severity (UPDRS) prediction under non-IID data environments.
- Developed adaptive aggregation using RMSE-based weighting, client-size normalization, and ensemble model fusion to improve global model performance across distributed clients.
- Integrated SHAP explainability, Kendall Tau consistency analysis, and statistical validation (Bootstrap CI, Wilcoxon testing), achieving RMSE of 4.51 and  $R^2$  of 0.8162 under decentralized, privacy-preserving constraints.

## TECHNICAL SKILLS

**Programming:** Python, C++

**Machine Learning & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, CNN, LSTM, GRU, Ensemble Learning, Time Series Forecasting

**Federated Learning & XAI:** Federated Learning, Explainable AI (SHAP), Adaptive Aggregation, Statistical Validation

**Generative AI:** LLMs, RAG, LangChain, LangGraph, Pinecone, Vector Databases, Prompt Engineering

**Backend & APIs:** FastAPI, Streamlit, REST APIs

**Data & Tools:** NumPy, Pandas, Matplotlib, Git, GitHub

## PROFESSIONAL EXPERIENCE

### Virtual Intern – Google Cloud Generative AI

Sep 2024 – Oct 2024

SmartBridge & SmartIntern (Supported by AICTE)

- Built Generative AI applications on Google Cloud, creating interactive demos using Streamlit for text generation and automation workflows.
- Designed end-to-end pipelines using prompt engineering techniques for real-world text generation tasks.

### AI/ML Virtual Internship

Jan 2025 – Mar 2025

EduSkills, Google for Developers, AICTE

- Implemented classification and prediction models using PyTorch/TensorFlow on real-world datasets with preprocessing and feature engineering.
- Built interactive model demos using **Streamlit** for visualizing predictions and evaluation metrics.

### ServiceNow Academic India Cohort

Mar 2025 – Aug 2025

ServiceNow

- Trained on ServiceNow ITSM platform covering workflow automation & enterprise IT applications.
- Acquired hands-on experience with cloud-based automation tools & IT service management concepts.

LEADERSHIP & ACHIEVEMENTS

Secretary, Microsoft Azure Community (MIET) – Coordinated Azure-related community activities and technical events.  
Graphic Designer Head, Society of Electronics Engineers (SEE) – Led graphics design team for technical and cultural events.  
Ranked in Top 15 among B.Tech CSE students in 1st and 2nd year academic assessments, MIET Meerut.

CERTIFICATIONS

- **Analyze Speech & Language with Google APIs** — Google Cloud | Oct 2025
- **Digital Application Fundamentals – STEM (Gold, 94.7%)** — FutureSkills Prime (NASSCOM, Govt. of India) | Mar 2025
- **Simulink Onramp** — MathWorks Training Services | May 2025
- **Introduction to Cybersecurity** — Cisco Networking Academy | Jan 2025

EDUCATION

|                                                                                                        |                        |
|--------------------------------------------------------------------------------------------------------|------------------------|
| <b>Dr. A.P.J. Abdul Kalam Technical University</b>                                                     |                        |
| <i>Bachelor of Technology in Computer Science &amp; Engineering (AI Specialization), CGPA: 9.02/10</i> | <i>Oct 2023 – 2027</i> |
| <b>DAV Public School (CBSE Board)</b>                                                                  |                        |
| Senior Secondary (XII)                                                                                 | 93.8%   2023           |
| Secondary (X)                                                                                          | 95.8%   2021           |